

From Quadratic Reciprocity, to Artin's Reciprocity, to Langlands's Reciprocity

This article assumes the reader has knowledge of some algebraic number theory, namely knowledge of concepts such as quadratic reciprocity, Frobenius automorphism, Galois groups, and similar concepts. The Kronecker-Weber Theorem will be mentioned in this article, so for the reader who is unaware of this result, it states that all abelian extensions of $\mathbb{Q}$ can be embedded in a Cyclotomic extension of $\mathbb{Q}$, so every finite abelian extension of $\mathbb{Q}$ is a subfield of some $\mathbb{Q}(\zeta_n)$. It is containment, not equality: $\mathbb{Q}(\sqrt{2})$ is abelian but is not itself cyclotomic.

When looking for solutions or irreducibility of integer polynomials, it is natural to want to find as much information as possible about the polynomial. One strategy is to find the solutions of a polynomial mod primes, and trying to piece together this information. As Galois groups hold information about the relation between the roots of polynomials, it would be nice if the “local” information provided by solving after modding can be linked to the “global” information provided by the Galois group. The Artin reciprocity can be interpreted as one part in achieving this result.

To demonstrate this, recall that p is a quadratic residue mod q ($n^2 \equiv p \pmod{q}$) if and only if $\left(\frac{p}{q}\right) = 1$ where $\left(\frac{p}{q}\right)$ is the Legendre symbol. This symbol satisfies the following equation:

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{q-1}{2} \frac{p-1}{2}}$$

This is known as the *Quadratic Reciprocity law*. Notice that we can consider the image of the Legendre symbol to be isomorphic to $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q}))$ (they are both $\cong \mathbb{Z}/2\mathbb{Z}$). The law involves two symbols, and it matters which one we draw: we put $\left(\frac{q}{p}\right)$ on the arrow, with q fixed and p varying over primes, not $\left(\frac{p}{q}\right)$; the computation below will show this choice is forced. It would be more interesting if there is a function from the primes of $\mathbb{Q}(\sqrt{q})/\mathbb{Q}$ to the Galois group that commutes with this symbol:

$$\begin{array}{ccc} \{\text{primes of } \mathbb{Q}(\sqrt{q})/\mathbb{Q}\} & \xrightarrow{\left(\frac{q}{p}\right)} & \{-1, 1\} \\ \downarrow \text{id} & & \downarrow \cong \\ \{\text{primes of } \mathbb{Q}(\sqrt{q})/\mathbb{Q}\} & \xrightarrow{?} & \text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q})) \end{array}$$

It would then be not too much a stretch to ask if we can replace the domain of these functions with the collection of all ideals of $\mathbb{Q}(\sqrt{q})/\mathbb{Q}$:

$$\begin{array}{ccc} \mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} & \xrightarrow{\Pi_p\left(\frac{q}{p}\right)} & \{-1, 1\} \\ \downarrow \text{id} & & \downarrow \cong \\ \mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} & \xrightarrow{?} & \text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q})) \end{array}$$

In fact, nothing at “?” needs to be quoted from elsewhere; it can be computed. Let p be an odd prime not dividing q and let $\mathfrak{p}$ be a prime above it. Frobenius acts on residues as $x \mapsto x^p$, so modulo $\mathfrak{p}$,

$$\text{Frob}_{\mathfrak{p}}(\sqrt{q}) \equiv (\sqrt{q})^p = q^{\frac{p-1}{2}} \sqrt{q} \equiv \left(\frac{q}{p}\right) \sqrt{q},$$

the last congruence being Euler's criterion. Since $\text{Frob}_{\mathfrak{p}}(\sqrt{q})$ is $\pm\sqrt{q}$ and the congruence fixes the sign, the bottom arrow is forced, and it depends only on p and not on the choice of $\mathfrak{p}$: it is the Legendre

symbol itself. The condition “ $\left(\frac{q}{p}\right) = 1$, i.e. q is a quadratic residue mod p ” becomes “ $\text{Frob}_p = \text{id}$ ”, i.e. “ p splits completely”:

$$\begin{array}{ccc} \mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} & \xrightarrow{\Pi_p\left(\frac{q}{p}\right)} & \{-1, 1\} \\ \downarrow = & & \downarrow \cong \\ \mathcal{I}_{\mathbb{Q}(\sqrt{q})/\mathbb{Q}} & \xrightarrow{\text{Frob}_p} & \text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{q})) \end{array}$$

The reader should watch this happen once; take $q = 5$. For $p = 11$: $5^5 = 3125 \equiv 1 \pmod{11}$, so $\text{Frob}_{11}(\sqrt{5}) = \sqrt{5}$, $\text{Frob}_{11} = \text{id}$, and 11 splits, as indeed $11 = (4 + \sqrt{5})(4 - \sqrt{5})$. For $p = 3$: $5 \equiv -1 \pmod{3}$, so $(\sqrt{5})^3 = 5\sqrt{5} \equiv -\sqrt{5}$, Frob_3 is the non-trivial automorphism, and 3 is inert. The congruence side agrees: $11 \equiv 1 \pmod{5}$ and 1 is a square mod 5, while 3 is not one (the squares mod 5 are 1 and 4). Reciprocity demands this agreement, since $5 \equiv 1 \pmod{4}$.

The choice of symbol deserves a pause, because the two sides of the law are different in kind. The computation shows that $\left(\frac{q}{p}\right)$ is the *Galois* side: it is defined by how p behaves in $\mathbb{Q}(\sqrt{q})$. The symbol $\left(\frac{p}{q}\right)$ is a congruence condition on p modulo q . They are not the same function of p : for $q = 3$ we have $\left(\frac{7}{3}\right) = 1$ yet 7 is inert, and $\left(\frac{11}{3}\right) = -1$ yet 11 splits; only $\left(\frac{q}{p}\right)$ tracks splitting, and the two agree for all p exactly when $q \equiv 1 \pmod{4}$. Quadratic reciprocity is the statement that the Galois side equals the congruence side, up to the sign factor when $p, q \equiv 3 \pmod{4}$. That the splitting of p in $\mathbb{Q}(\sqrt{q})$ is decided by p modulo q is the surprise, and it is what makes the diagram worth drawing.

As nice as an observation this is, as we are working with such small sets, the question becomes whether this is an accident or a pattern; can $\text{Gal}_{\mathbb{Q}}(K)$ (or even $\text{Gal}_K(L)$) be replaced with another abelian Galois group, where the abelian condition is necessary for the Frobenius automorphism to be well-defined, and we can find a corresponding “Legendre symbol” and a group in the codomain that would make the above diagram commute? Does this generalize when looking at cubic reciprocity, biquadratic reciprocity, and so forth (with the appropriate replacement of the Legendre symbol and its image)? Phrased differently, for different polynomial equations, when taking the mod by different primes and finding the relations between primes given by the polynomial, are these relations related to the Galois group? This was a major endeavour in the 20th century, notably Hilbert’s 23rd problem was:

Find a proof of the most general reciprocity law for an arbitrary number field

As it turns out, such a law exists! The group that replaces $\{-1, 1\}$ will be the *ideal class group* $\text{Cl}(K)$ (or more specifically, the *ray class group*)¹, and the Legendre symbol shall be generalized to the Artin symbol! The result is called *Artin Reciprocity*. Exactly how this replacement happens is the goal of a first course in class field theory, and so hopefully this is enough to convince the reader it is a topic worth learning.

One interesting result from this generalization is that most groups will not be isomorphic to the class group, but the *quotient* of the class group. The reader may rightfully wonder if there is a maximal extension L/K where $\text{Gal}_K(L)$ is isomorphic to the class group. As it will again turn out, the answer is yes and results from a generalization of Kronecker Weber’s theorem! Thus, the commutative diagram presented above would be extended to a commutative diagram:

$$\begin{array}{ccc} \{\text{primes of } K\} & \longrightarrow & \text{Gal}_K(L) \\ \downarrow \text{id} & & \downarrow \cong \\ \{\text{primes of } K\} & \longrightarrow & \text{Cl}(K)/H \end{array}$$

¹The ray class group refines the ideal class group rather than coarsening it: one fixes a modulus $\mathfrak{m}$ and quotients the ideals coprime to $\mathfrak{m}$ by the principal ideals having a generator congruent to 1 modulo $\mathfrak{m}$, which adds congruence and sign conditions at the places dividing $\mathfrak{m}$. If we considered the ideal class group, we get cases where the ideal class group is trivial while the Galois group is non-trivial, ex. consider $x^4 + 1$

and if $L = K_H$ is an appropriate “maximal” abelian field extension, then:

$$\mathrm{Cl}(K) \cong \mathrm{Gal}_K(K_H)$$

This incredible unification of the properties of primes and the polynomial relationships that can be put to associate them can be thought of as a crowning achievement of number theory in the 20th century.

One important caveat that must be mentioned is that this reciprocity works with *abelian* Galois groups. Intuitively, the reader can think of this as a consequence of the ideal group being abelian, having its multiplication structure on principal ideal mirror that of the multiplication structure from the group of units of a field. Is there an interesting suitable replacement or generalization of the ideal group or class groups that will allow for the reciprocity to generalize to all Galois groups? Such questions are the beginnings of Langlands’s program.

Before going there, let us see exactly where the abelian hypothesis bites. Fix a Galois extension L/K and a prime p of K unramified in L . For each prime $\mathfrak{P}$ of L above p there is a unique $\mathrm{Frob}_{\mathfrak{P}} \in \mathrm{Gal}_K(L)$ acting as $x \mapsto x^p$ on the residue field (see EYNTKA Algebra). But $\mathrm{Gal}_K(L)$ permutes the primes above p transitively, with $\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma \mathrm{Frob}_{\mathfrak{P}} \sigma^{-1}$, so p determines a *conjugacy class*, not an element. The class is a singleton for every p exactly when the group is abelian; for $x^3 - 2$, whose Galois group is S_3 , the three transpositions form a single class, and there is no way to choose among them. The Artin map sends an ideal to a group element, so no such map can exist non-abelianly; the obstruction is structural, not technical. There is, however, exactly enough data left: a conjugacy class is not an element, but any *class function* can be evaluated on it, and characters of representations are class functions. So while Frob_p is not an element, $\mathrm{tr} \rho(\mathrm{Frob}_p)$ is a well-defined number for every representation ρ of $\mathrm{Gal}_K(L)$ and every unramified p , abelian or not. This is why the non-abelian theory will be phrased in terms of representations: it is the smallest change that lets the data survive.

Towards Langlands’s Reciprocity

In the process of developing class field theory, it will come out that there are two approaches to the problem: one done by focusing on primes along with “infinite primes” (i.e. places) which leads to the ray class group. The other focuses on *local fields* and putting together the information of local fields to find global results. This approach is preferable to the approach given by Ray class groups as it allows us to consider the behaviour of all local fields, and hence all primes, at once. Considering the behaviour of all the primes at once can be thought of as the starting point for looking into objects should replace the class groups in the generalization of Artin’s Reciprocity.

An incredible observation in the 20th century was that we can capture the information given by primes by a generating function, and these shall be the appropriate replacement. To illuminate a bit with a special case, take a subset of polynomials $p(x, y)$ that are elliptic curves (polynomials of the form $y^2 = x^3 + ax + b$). We can take the solution of such a polynomial mod primes, count the number of solutions, normalize them², and make them the coefficients of a generating function. For example, for the elliptic curve $y^2 = x^3 - x$ is:

$$f(q) = q - 2q^5 - 3q^9 + 6q^{13} + 2q^{17} - q^{25} - 10q^{29} - 2q^{37} + \dots$$

We can compute one of these coefficients. Mod 5, the polynomial $x^3 - x$ takes the values 0, 0, 1, 4, 0 at $x = 0, 1, 2, 3, 4$, and the nonzero squares mod 5 are 1 and 4. So the solutions are (0, 0), (1, 0), (4, 0), then (2, 1), (2, 4) from $y^2 = 1$ and (3, 2), (3, 3) from $y^2 = 4$: seven points, and $a_5 = 5 - 7 = -2$, the printed coefficient of q^5 . The gap before it is no accident: mod 3 every x satisfies $x^3 \equiv x$, so there are three solutions and $a_3 = 3 - 3 = 0$, which is why nothing appears at q^3 . The same count at $p = 13$ gives seven points again and $a_{13} = 6$. Even the composite coefficients carry no new information: these coefficients satisfy $a_{p^2} = a_p^2 - p$ (the form has weight 2), so $a_9 = 0 - 3 = -3$ and $a_{25} = 4 - 5 = -1$, both as printed. And there is more in a_p than a normalized count: a_p is the trace of Frob_p acting on a two-dimensional Galois representation attached to the curve, its ℓ -adic Tate module. So $a_5 = -2$ is the trace of a Frobenius

²In particular, take $p - \# \text{Sol}$ to normalize the solution-set so that these hover around 0

conjugacy class, exactly the quantity that survives when the group is non-abelian, and the series above is a list of such traces; that this list has the symmetries described next is the two-dimensional reciprocity this section is heading towards. This generating function will exhibit incredible symmetries, namely it is invariant under certain group actions³. Generating functions that exhibit these nice properties are called *modular form*. These are good candidates for replacing the class group. Furthermore, instead of working directly with non-abelian groups, we shall work with Galois representations. One-dimensional representations correspond to the abelianised Galois group and recover Artin Reciprocity.

One of the great achievements of the end of the 20th century settled an important two-dimensional case! These correspond to elliptic curves, and it was shown that all elliptic curves correspond to a modular form, that is, all elliptic curves are *modular*⁴. One consequence of this is that Fermat's last theorem was shown to be true: that for $n \geq 3$ the equation $x^n + y^n = z^n$ has no solution in nonzero integers. This is because this statement can be shown to be equivalent to the following:

If $a, b, c \in \mathbb{N}$ is a non-trivial solution to $a^p + b^p = c^p$ for an odd prime p , then

$$y^2 = x(x - a^p)(x + b^p)$$

will be an elliptic curve without a modular form

This implication is Ribet's theorem. Since every elliptic curve over $\mathbb{Q}$ does have a modular form, no such curve can exist, and therefore no such solution a, b, c can exist either.

The hunt is now on for pushing further the Langlands Correspondence! Over function fields, which behave much like number fields, the correspondence was established by Drinfeld and Lafforgue; and the *Geometric Langlands correspondence*, which works instead with sheaves on a curve over $\mathbb{C}$, was recently proven. There is still a lot of work for the higher-dimensional versions of Langlands's correspondence, and though daunting to understand, it is fascinating to see it come together!

³This symmetry can be seen by writing this function in terms of eta functions, something that would be covered in great depth while studying modular forms

⁴Wiles, in the paper that proved Fermat's Last Theorem, showed that all semi-stable elliptic curves over $\mathbb{Q}$ are modular; Breuil, Conrad, Diamond and Taylor extended this to all elliptic curves over $\mathbb{Q}$ a few years later